Expt: 8203
Date:

STUDY OF NETWORK CABLES

1) Understand different types of network cables

→ AIM: Study of different types of Network cables - understand different types of network cables.

→ 1) Different types of network cable used in networking are:
   i) Unshielded Twisted Pair (UTP) Cable
   ii) Sheilded Twisted Pair (STP) cable.
   iii) Coaxial cable
   iv) Fibre optic cable.

CABLE TYPE	CATEGORY	MAXIMUM DATA TRANSMISSION	ADVANTAGES / DISADVANTAGES	APPLICATION / USE
UTP	category 3	10 bps	advantages • Cheaper in cost • Easy to install disadvantages • More prone to (EMI) Electromagnetic interference & noise.	10Base-T Ethernet
	category 5	Upto 100 Mbps		Fast Ethernet, Gigabit Ethernet
	category 5e	1Gbps		Fast Ethernet Gigabit Ethernet.
STP	Category 6, 6a	10Gbps	advantages • Sheilded • Faster than UTP • Less susceptible to noise & interference. disadvantages • Expensive • Greater installation effort.	Gigabit Ethernet, 10G Ethernet (55m) widely used in data centres.
SSTP	Category 7	10Gbps	" "	Gigabit Ethernet, 10G Ethernet (100m)

Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	• High bandwidth • Immune to interference. • Low loss bandwidth • Versatile. disadvantage: • Limited distance • cost • Size is bulky	Speed of signal is 500m Television. High speed internet.
Fibre optics cable	Single mode, Multi mode	100 Gbps	advantages • High speed • High bandwidth • High security disadvantages • Expensive • Requires skilled installes	Maximum distance of fibre optic cable is around 100 meters.

b) Make an Ethernet cross-over cable:

Tools:

→ Ethernet cabling
→ A crimpling tool
→ Two RJ45 plugs
→ Two plug shields

Step 1:

Begin by threading sheilds on to the cable.

Step 2:

Next, strip approx. 1-5 cm of cable sheilding from both ends

Step 3:

Untangle the wires, there should be four twisted pairs. One end should be four twisted pairs.

Step-4:

Bunch them together in a line, stick out farther than others, ship them back, Hold the plug with clip side facing away from you & pins towards you

Step-5:

The notch at the end of the plug needs to be just over cable shielding.

Step-6:

After securing the plug, insert into cramping tool.

Step-7:

Lastly repeat for other end with B.

RESULT:

Thus, the study of Network cables (a, b) have been executed & verified successfully.